SBT-DF204 — Computer Forensics Case Study

Case Study 3 Individual memory forensics investigation

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Course: SBT-DF204-Computer Forensics Case Study

Lab Title: Investigating Suspected Illegal File Transfer From Memory Evidence

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Date: 19/10/2026

2. Executive Conclusion

The memory forensic investigation of `memdumpWin7.mem` identified multiple artifacts consistent with the copying of a file named `secret_file.docx` from the Downloads folder to a removable-storage drive identified as `F:`.

Volatility Framework 2.6.1 was used with the selected profile `Win7SP1x86_23418`. The recovered console history contains the command "copy secret_file.docx F, and the console screen buffer displays the result `1 file(s) copied.` Process analysis identified `cmd.exe` (PID 3996) and its associated console process, `conhost.exe` (PID 3988), both created at 2019-01-06 15:06:11 UTC. This is the process creation time, not the confirmed execution time of the copy command.

Additional registry evidence identified a General UDisk USB mass-storage device and associated the device with drive letter `F. MountPoints2 and Shellbags artifacts provide further context linking Windows Explorer activity with the Downloads folder and the `F drive. These independent artifacts collectively support the conclusion that a copy operation to a USB-associated destination was reported by the Windows command console.

The investigation does not establish the exact time the file was copied, independently verify the file's contents or integrity on the destination device, identify the individual who operated the computer, or determine whether the transfer was authorized or unlawful. The findings therefore support a reported file-copy operation to a removable-storage-associated drive, but do not, by themselves, prove illegal data exfiltration.

The original and working memory images had matching SHA-256 hashes of 500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e, confirming that the two images were identical when hashed. This supports the integrity of the working copy relative to the original image provided for analysis. The final assessment remains subject to the limitations of the available memory snapshot and the artifacts recoverable from it.

3. Evidence acquisition and integrity

Item	**Record**
Source of memory image	Course Materials
Original file name	memdumpWin7.mem
File size (bytes)	1073676288
SHA-256 (original)	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
SHA-256 (working copy)	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
Working-copy method	cp from the read-only original, then hash comparison
File-access precautions	Original not opened by any tool, all analysis on the working copy, hashes recorded before analysis
Hash re-check at end	

The supplied memory image was obtained from the course materials and identified as memdumpWin7.mem. The original image has a size of 1,073,676,288 bytes and a SHA-256 hash of 500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e. A working copy was created using cp, and its SHA-256 hash matched the original. The original image was not used directly for analysis; examination was performed on the working copy. A final hash re-check will be performed after the forensic examination to confirm that the working copy remained unchanged.

4. Method and profile selection

4.1 Tools and environment

Item	Record
Operating environment (e.g. ICDFA Kali lab)	Kali GNU/Linux Rolling 2026.3, kernel Linux 7.1.5+kali-amd64 (x86_64).
Volatility version	Volatility Foundation Volatility Framework 2.6.1, cloned from github.com/volatilityfoundation/volatility, commit 328a178edeecaf8d2abf96618ecfcb14a7b0c860 (dated 16 May 2025)
Launch command	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 <plugin>, run from ~/volatility
Other tools and versions	Python 2.7.18; pip 20.3.4; pycryptodome 3.24.1; git 2.53.0; sha256sum (GNU coreutils 9.10); grep and awk (add their versions if your lecturer wants them)
Variations from brief commands (and why)	The six points from my last message, edited to match what you actually did (manual install, pycryptodome added, distorm3 skipped, output filtering, working copy only, re-download)

The memory image was examined using Volatility 2 in the authorised Kali laboratory environment. The imageinfo plugin was first used to identify candidate Windows profiles and determine the image time. Win7SP1x86_23418 was selected because it was ranked first by imageinfo and produced coherent output during subsequent examination. The profile selection was treated cautiously because pslist and hivelist results were identical for three candidate profiles, meaning those plugins did not independently distinguish the candidates. The Win7SP1x86 candidate was not separately tested and therefore is not presented as validated by those plugins.

4.2 Imageinfo result and profile selection

Item	Evidence
Suggested profiles	Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
Selected profile	Win7SP1x86_23418
Reason	Ranked first by imageinfo, Image Type SP1, and pslist, netscan and hivelist returned coherent output
Validation plugins and results	pslist rows and hivelist hash (8fd9933b...) identical for three candidates, so these plugins

	don’t discriminate between them. I haven’t tested Win7SP1x86 separately, so don’t claim it was.
Image date and time (UTC)	2019-01-06 15:09:06
Time zone shown	UTC−8, registry says Pacific Standard Time
Local-time treatment	UTC is primary, local shown separately as PST

The memory image time reported by the examination was 2019-01-06 15:09:06 UTC. The associated timezone information indicated UTC−8/Pacific Standard Time. UTC is therefore used as the primary reference for the investigation timeline, with local PST time shown separately where necessary.

4.3 Plugin and command register

Ref	Plugin	Full command run	Output file	Figure ref
V1	imageinfo	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem imageinfo	V1_imageinfo.txt	Figure 1
V2	pslist	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 pslist	V2_pslist.txt	Figure 2
V3	psscan	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 psscan	V3_psscan.txt	Figure 3
V4	netscan	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 netscan	V4_netscan.txt	Figure 4
V5	cmdscan	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 cmdscan	V5_cmdscan.txt*	Figure 5
V6	consoles	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 consoles	V6_consoles.txt*	Figure 6
V7	hivelist	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 hivelist	V7_hivelist.txt	Figure 7

V8	printkey (USBSTOR)	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\\Enum\\USBSTOR"	V8_USBSTOR.txt *	Figure 8
V9	printkey (MountPoints2)	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 printkey -K "Software\\Microsoft\\Windows\\CurrentVersion\\Explorer\\MountPoints2"	V9_MountPoints2.txt*	Figure 9
V10	shellbags	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 shellbags	V10_shellbags.txt *	Figure 10
V11	printkey (ProfileList)	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 printkey -K "Microsoft\\Windows NT\\CurrentVersion\\ProfileList"	V11_ProfileList.txt	Figure 11

5. Findings

Q1. Evidence image acquired and preservation
Observed fact / evidence IDs

The original memory image was stored at /home/kali/mem_lab/original/memdumpWin7.mem, and the working copy was stored at /home/kali/mem_lab/working/memdumpWin7_working.mem. Both files had a reported size of 1,073,676,288 bytes. Their SHA-256 hashes matched:

500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e

```
┌──(kali㉿kali)-[~/mem_lab/original]
└─$ sha256sum ~/mem_lab/original/memdumpWin7.mem
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  /home/kali/mem_lab/original/mem
dumpWin7.mem

┌──(kali㉿kali)-[~/mem_lab/original]
└─$ chmod 444 ~/mem_lab/original/memdumpWin7.mem
cp ~/mem_lab/original/memdumpWin7.mem ~/mem_lab/working/memdumpWin7_working.mem
sha256sum ~/mem_lab/working/memdumpWin7_working.mem
date -u
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  /home/kali/mem_lab/working/memd
umpWin7_working.mem
Mon Oct 19 08:10:18 AM UTC 2026
```

Interpretation

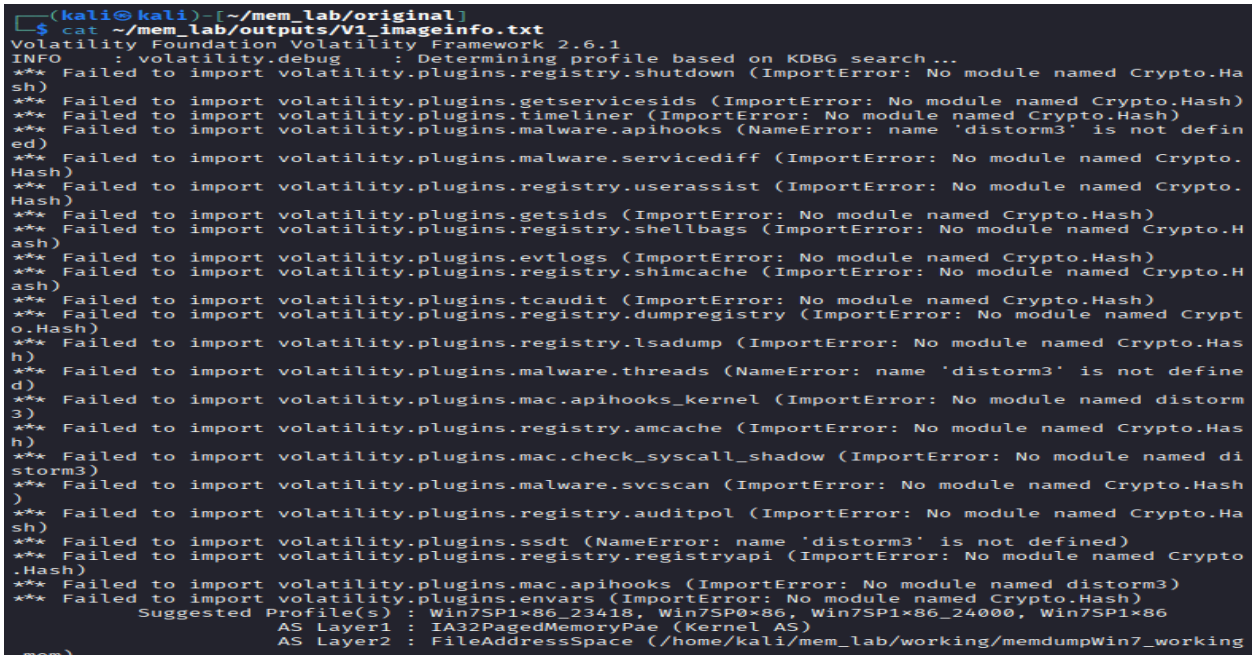
The matching file sizes and SHA-256 hashes support the conclusion that the original and working copies were byte-for-byte identical when the hashes were calculated. Analysis was conducted using the working copy to reduce the risk of altering the original evidence.

Limit

Matching hashes establish that the two files matched at the time of verification. They do not independently establish the original image's acquisition method, source authenticity, or complete chain of custody.

Q2. Operating-system profile and image time supported
Observed fact / evidence IDs

Volatility Framework 2.6.1 imageinfo suggested Win7SP1x86_23418 as its first profile candidate. The output identified Windows 7 Service Pack 1 and reported the image time as 2019-01-06 15:09:06 UTC. The selected profile was Win7SP1x86_23418.



Interpretation

The imageinfo results support using Win7SP1x86_23418 as the initial analysis profile. The reported timestamp provides a reference time for the memory image and for interpreting any timestamps recovered from it.

Limit

The suggested profile is not, by itself, conclusive proof of the operating-system build. The image timestamp is not necessarily the time of the suspected file transfer. Plugin compatibility and additional artefacts should be considered when validating the profile.

Q3. User and account artefacts (SIDs, hive paths, profile vs person)
Observed fact / evidence IDs

```
┌──(kali㉿kali)-[~/mem_lab/original]
└─$ cat ~/mem_lab/outputs/V8_profilelist.txt
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

**********
Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: ProfileList (S)
Last updated: 2015-09-21 09:50:52 UTC+0000

Subkeys:
  (S) S-1-5-18
  (S) S-1-5-19
  (S) S-1-5-20
  (S) S-1-5-21-1716914095-909560446-1177810406-1000
  (S) S-1-5-21-1716914095-909560446-1177810406-1002

Values:
REG_EXPAND_SZ ProfilesDirectory : (S) %SystemDrive%\Users
REG_EXPAND_SZ Default           : (S) %SystemDrive%\Users\Default
REG_EXPAND_SZ Public            : (S) %SystemDrive%\Users\Public
REG_EXPAND_SZ ProgramData       : (S) %SystemDrive%\ProgramData
```

Interpretation

The SID entries ending in -1000 and -1002 are candidate user-account profile identifiers for further investigation. The profile registry information helps establish where Windows user profiles are stored.

Limit

A SID or profile entry does not, by itself, establish the real-world identity of the person using the account. The available results do not yet confirm which account, if either, was associated with the recovered copy command.

Q4. Process and network evidence (PIDs, endpoints, connection state)
Observed fact / evidence IDs

```
┌──(kali㉿kali)-[~/mem_lab/original]
└─$ cat ~/mem_lab/outputs/V2_pslist.txt
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defin
ed)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not define
d)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm
3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named di
storm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start
          Exit
---------- -------------------- ------ ------ ------ -------- ------ ------ ------------------------
---------- ------------------------------

0x8413a908 System                    4      0     73      518 ------      0 2019-01-06 18:02:44 U
TC+0000
0x851ec488 smss.exe                236      4      2       29 ------      0 2019-01-06 18:02:44 U
TC+0000
0x852b11e0 csrss.exe               300    292      8      434      0      0 2019-01-06 18:02:44 U
TC+0000
0x841c32b8 wininit.exe             340    292      3       75      0      0 2019-01-06 18:02:44 U
TC+0000
0x852b21d0 csrss.exe               348    328      9      316      1      0 2019-01-06 18:02:44 U
TC+0000
0x858a4d28 winlogon.exe            388    328      3      112      1      0 2019-01-06 18:02:45 U
TC+0000
0x858b0310 services.exe            432    340     10      228      0      0 2019-01-06 18:02:45 U
TC+0000
```

```
┌──(kali㉿kali)-[~/mem_lab/original]
└─$ cat ~/mem_lab/outputs/V3_psscan.txt
Volatility Foundation Volatility Framework 2.6.1
Offset(P)          Name                PID   PPID PDB        Time created                   Time
exited
------------------ ---------------- ------ ------ ---------- ------------------------------ ----
------------------------

0x000000003e0287f0 WUDFHost.exe       3300    828 0x3efef4c0 2019-01-06 15:03:07 UTC+0000

0x000000003e27f030 VBoxTray.exe       2632   2524 0x3efef560 2019-01-06 15:02:55 UTC+0000

0x000000003e2b9820 SearchIndexer.     2960    432 0x3efef580 2019-01-06 15:03:01 UTC+0000

0x000000003e407030 svchost.exe        2072    432 0x3efef520 2019-01-06 15:02:50 UTC+0000

0x000000003e423ce8 svchost.exe        1064    432 0x3efef220 2019-01-06 15:02:48 UTC+0000

0x000000003e44fd28 spoolsv.exe        1184    432 0x3efef240 2019-01-06 15:02:48 UTC+0000

0x000000003e491030 svchost.exe        1228    432 0x3efef260 2019-01-06 15:02:48 UTC+0000

0x000000003e4de488 vmicsvc.exe        1364    432 0x3efef2a0 2019-01-06 15:02:48 UTC+0000

0x000000003e4e3030 vmicsvc.exe        1388    432 0x3efef2c0 2019-01-06 15:02:48 UTC+0000

0x000000003e4e9030 taskhost.exe       1412    432 0x3efef2e0 2019-01-06 15:02:48 UTC+0000

0x000000003e4f6d28 vmicsvc.exe        1440    432 0x3efef300 2019-01-06 15:02:48 UTC+0000

0x000000003e506d28 vmicsvc.exe        1492    432 0x3efef340 2019-01-06 15:02:48 UTC+0000
```

```
┌──(kali㉿kali)-[~/mem_lab/original]
└─$ cat ~/mem_lab/outputs/V4_netscan.txt
Volatility Foundation Volatility Framework 2.6.1
Offset(P)          Proto    Local Address                  Foreign Address      State
Pid      Owner          Created
0x21478bb0         TCPv4    0.0.0.0:49155                  0.0.0.0:0            LISTENING
432      services.exe
0x269ff708         TCPv4    0.0.0.0:49156                  0.0.0.0:0            LISTENING
440      lsass.exe
0x269ff708         TCPv6    :::49156                       :::0                 LISTENING
440      lsass.exe
0x3e081f50         UDPv4    192.168.56.8:56647             *:*
2388     chrome.exe     2019-01-06 15:08:32 UTC+0000
0x3e2863d8         UDPv4    0.0.0.0:0                      *:*
616      VBoxService.ex 2019-01-06 15:08:35 UTC+0000
0x3e2a2480         UDPv6    fe80::1571:7b34:b2b1:bbe4:1900 *:*
2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e46edf0         UDPv4    192.168.56.8:137               *:*
4        System         2019-01-06 15:02:48 UTC+0000
0x3e46ef50         UDPv4    192.168.56.8:138               *:*
4        System         2019-01-06 15:02:48 UTC+0000
0x3e5c0f50         UDPv4    0.0.0.0:0                      *:*
2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv4    0.0.0.0:0                      *:*
2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv6    :::0                           *:*
2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c5910         UDPv4    0.0.0.0:5355                   *:*
1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv4    0.0.0.0:5355                   *:*
1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv6    :::5355                        *:*
1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e6074e0         UDPv6    ::1:51122                      *:*
2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e7f2828         UDPv4    0.0.0.0:0                      *:*
1064     svchost.exe    2019-01-06 15:02:49 UTC+0000
0x3e7f2828         UDPv6    :::0                           *:*
```

Interpretation

The process and network artefacts may help identify running processes and potential network activity at the time represented by the memory image. The recovered console history provides a process context for investigating the file-copy command.

Limit

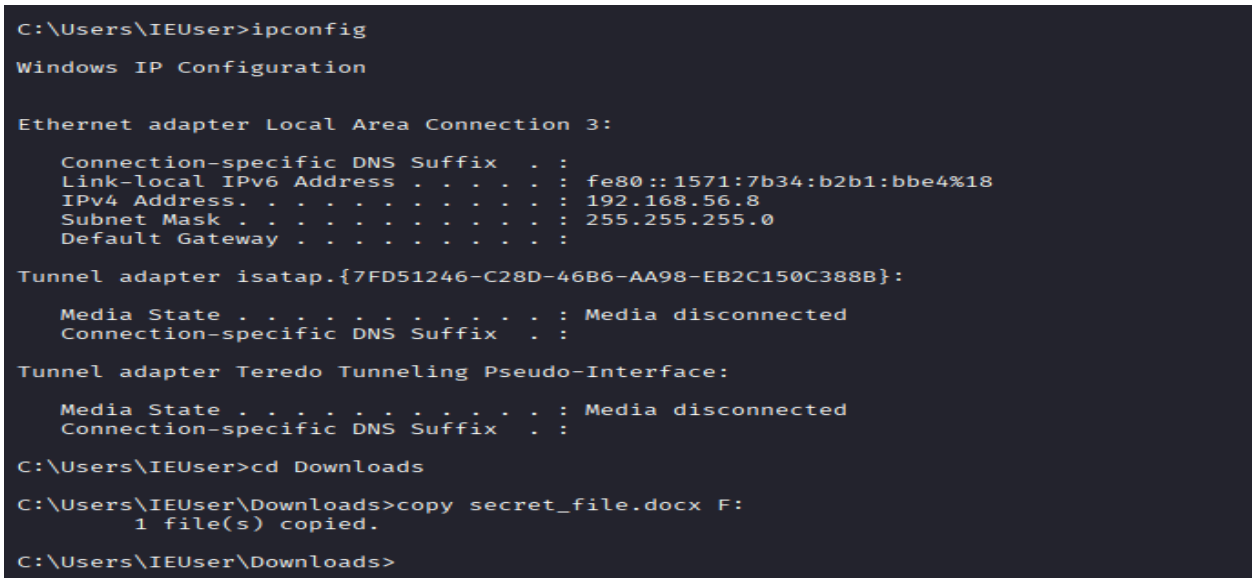
The specific process names, PIDs, endpoints, and connection states from the saved process and network outputs must be reviewed before drawing further conclusions. A network artefact does not automatically establish a successful connection or file transfer. No contact with recovered endpoints was performed as part of this analysis.

Q5. Command history (cmdscan / consoles)
Observed fact / evidence IDs

```
┌──(kali㉿kali)-[~/volatility]
└─$ python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem \
--profile=Win7SP1x86_23418 cmdscan
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defin
ed)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not define
d)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm
3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named di
storm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
**************************************************
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: 3?4?2????2
Cmd #37 @ 0x3200c4: 3?4?2????2
**************************************************
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????
Cmd #37 @ 0x10cff0: ?????
```

```
┌──(kali㉿kali)-[~/volatility]
└─$ cat ~/mem_lab/outputs/V6_consoles.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defin
ed)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not define
d)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm
3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named di
storm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
**************************************************
ConsoleProcess: conhost.exe Pid: 2000
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
Title: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
AttachedProcess: sshd.exe Pid: 2016 Handle: 0x54
----
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
----
CommandHistory: 0x3207f0 Application: - Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
----
Screen 0x3360a0 X:80 Y:300
Dump:

**************************************************
ConsoleProcess: conhost.exe Pid: 3988
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: %SystemRoot%\system32\cmd.exe
Title: C:\Windows\system32\cmd.exe
AttachedProcess: cmd.exe Pid: 3996 Handle: 0x5c
----
CommandHistory: 0x1105d8 Application: ipconfig.exe Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
```



Interpretation

The recovered console screen buffer provides evidence that the Windows command interpreter displayed a successful file-copy message for secret_file.docx, with F: specified as the destination. This is stronger evidence than the command-history entry alone and supports the conclusion that the copy operation was reported as successful by the command interpreter.

Limit

The recovered console output does not independently establish the physical identity of drive F, whether it was removable media, or whether the copied file remained intact at the destination. The output also does not establish the real-world identity of the person operating the account. The command has no verified execution timestamp in the recovered text shown here.

Q6. Removable-media evidence (USBSTOR, DeviceClasses, MountedDevices, MountPoints2)
Observed fact / evidence IDs

```
┌──(kali㉿kali)-[~/volatility]
└─$ cd ~/volatility

python2 vol.py \
-f ~/mem_lab/working/memdumpWin7_working.mem \
--profile=Win7SP1x86_23418 \
printkey -K 'ControlSet001\Enum\USBSTOR' \
2>&1 | tee ~/mem_lab/outputs/V12_USBSTOR.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defin
ed)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not define
d)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm
3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named di
storm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Legend: (S) = Stable   (V) = Volatile

----------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: USBSTOR (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
  (S) Disk&Ven_General&Prod_UDisk&Rev_5.00

Values:
```

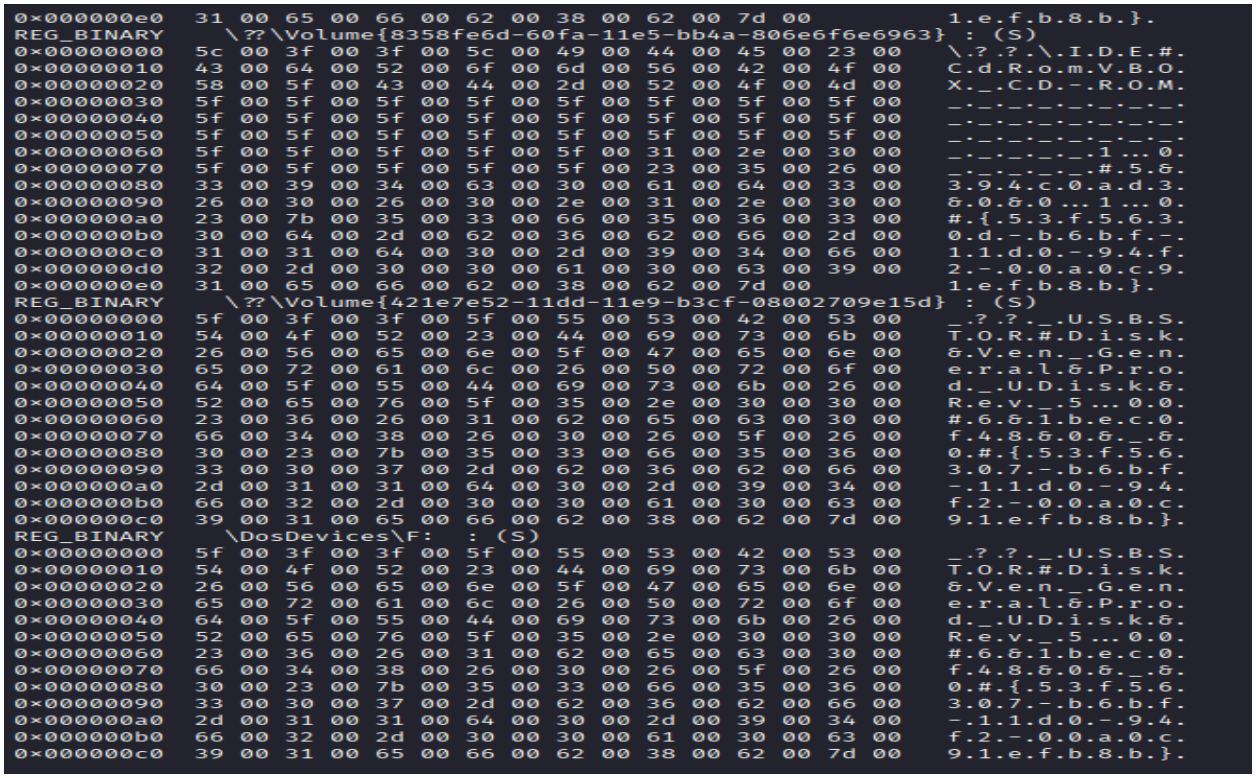
```
┌──(kali㉿kali)-[~/volatility]
└─$ python2 vol.py \
-f ~/mem_lab/working/memdumpWin7_working.mem \
--profile=Win7SP1x86_23418 \
printkey -K 'MountedDevices' \
2>&1 | tee ~/mem_lab/outputs/V14_MountedDevices.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defin
ed)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not define
d)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm
3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named di
storm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Legend: (S) = Stable   (V) = Volatile

----------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: MountedDevices (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

Values:
REG_BINARY     \DosDevices\C:  : (S)
0x00000000  cd d7 ee 50 00 00 10 00 00 00 00 00                 ...P........
REG_BINARY     \??\Volume{a5b8a980-608c-11e5-a266-806e6f6e6963} : (S)
0x00000000  cd d7 ee 50 00 00 10 00 00 00 00 00                 ...P........
REG_BINARY     \DosDevices\D:  : (S)

REG_BINARY     \DosDevices\E:  : (S)
0x00000000  5c 00 3f 00 3f 00 5c 00 49 00 44 00 45 00 23 00    \.?.?.\.I.D.E.#.
0x00000010  43 00 64 00 52 00 6f 00 6d 00 56 00 42 00 4f 00    C.d.R.o.m.V.B.O.
0x00000020  58 00 5f 00 43 00 44 00 2d 00 52 00 4f 00 4d 00    X._.C.D.-.R.O.M.
0x00000030  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00    _._._._._._._._.
0x00000040  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00    _._._._._._._._.
0x00000050  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00    _._._._._._._._.
0x00000060  5f 00 5f 00 5f 00 5f 00 5f 00 31 00 2e 00 30 00    _._._._._._1...0.
0x00000070  5f 00 5f 00 5f 00 5f 00 5f 00 23 00 35 00 26 00    _._._._._.#.5.&.
0x00000080  33 00 39 00 34 00 63 00 30 00 61 00 64 00 33 00    3.9.4.c.0.a.d.3.
0x00000090  26 00 30 00 26 00 30 00 2e 00 31 00 2e 00 30 00    &.0.&.0...1...0.
0x000000a0  23 00 7b 00 35 00 33 00 66 00 35 00 36 00 33 00    #.{.5.3.f.5.6.3.
0x000000b0  30 00 64 00 2d 00 62 00 36 00 62 00 66 00 2d 00    0.d.-.b.6.b.f.-.
0x000000c0  31 00 31 00 64 00 30 00 2d 00 39 00 34 00 66 00    1.1.d.0.-.9.4.f.
```



Interpretation

The recovered registry entry is consistent with a USB mass-storage device being represented in the Windows registry. In combination with the console command copy secret_file.docx F: and the displayed message 1 file(s) copied., this provides a reason to investigate whether the copy destination was associated with removable storage.

Limit

The USBSTOR entry alone does not establish which drive letter was assigned to the device or prove that this specific file was copied to it. The registry key's last-updated time is not necessarily the exact time the device was connected or the file was copied. Additional registry evidence is required to associate the device with F

Q7. File Explorer evidence (ShellBags, folder paths, access or preference times)

```
┌──(kali㉿kali)-[~/volatility]
└─$ cat ~/mem_lab/outputs/V17_shellbags.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defin
ed)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not define
d)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm
3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named di
storm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Scanning for registries....
Gathering shellbag items and building path tree ...
**************************************************************************
Registry: \??\C:\Users\IEUser\ntuser.dat
Key: Software\Microsoft\Windows\Shell\Bags\1\Desktop
Last updated: 2019-01-06 14:32:53 UTC+0000
Value                File Name       Modified Date                  Create Date
      Access Date                 File Attr                Unicode Name
-----  -------------------  ---------  -----------------------  -----------  ----------------------
ItemPos944×959×96(1)       GOOGLE~1.LNK    2018-09-17 14:55:16 UTC+0000   2018-09-17 14:53:04 UTC+0
000   2018-09-17 14:53:04 UTC+0000   ARC                              Google Chrome.lnk
ItemPos944×959×96(1)       eula.lnk        2015-09-21 09:19:50 UTC+0000   2015-09-21 09:19:50 UTC+0
000   2015-09-21 09:19:50 UTC+0000   ARC                              eula.lnk
ItemPos944×959×96(1)       OSFORE~1.LNK    2018-09-17 14:56:30 UTC+0000   2018-09-17 14:56:30 UTC+0
000   2018-09-17 14:56:30 UTC+0000   ARC                              OSForensics.lnk
ItemPos1024×768×96(1)      eula.lnk        2015-09-21 09:19:50 UTC+0000   2015-09-21 09:19:50 UTC+0
000   2015-09-21 09:19:50 UTC+0000   ARC                              eula.lnk
**************************************************************************

**************************************************************************
Registry: \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU
Last updated: 2019-01-06 15:04:40 UTC+0000
Value   Mru   Entry Type     GUID                                    GUID Description     Folder
 IDs
------  ----  -------------  --------------------------------------  -------------------  ------
----
1       4     Folder Entry   26ee0668-a00a-44d7-9371-beb064c98683    {Unknown CSIDL}      EXPLOR
ER, MY_COMPUTER, RECYCLE_BIN, UKNOWN
0       0     Folder Entry   20d04fe0-3aea-1069-a2d8-08002b30309d    My Computer          EXPLOR
ER, MY_COMPUTER
3       3     Folder Entry   f02c1a0d-be21-4350-88b0-7367fc96ef3c    Network              EXPLOR
```

Interpretation

These records support that Windows Explorer stored information about the Downloads folder and the F: volume in the user's Shellbags data. This is consistent with the recovered command history showing copy secret_file.docx F: and the USB registry evidence associating F: with a USB storage device.

Limit

Shellbags record folder-navigation and view-related information; they do not independently prove that the file was copied or opened. The recorded timestamps are not confirmed file-transfer times and do not establish who performed the operation or their intent.

Q8. Defensible conclusion

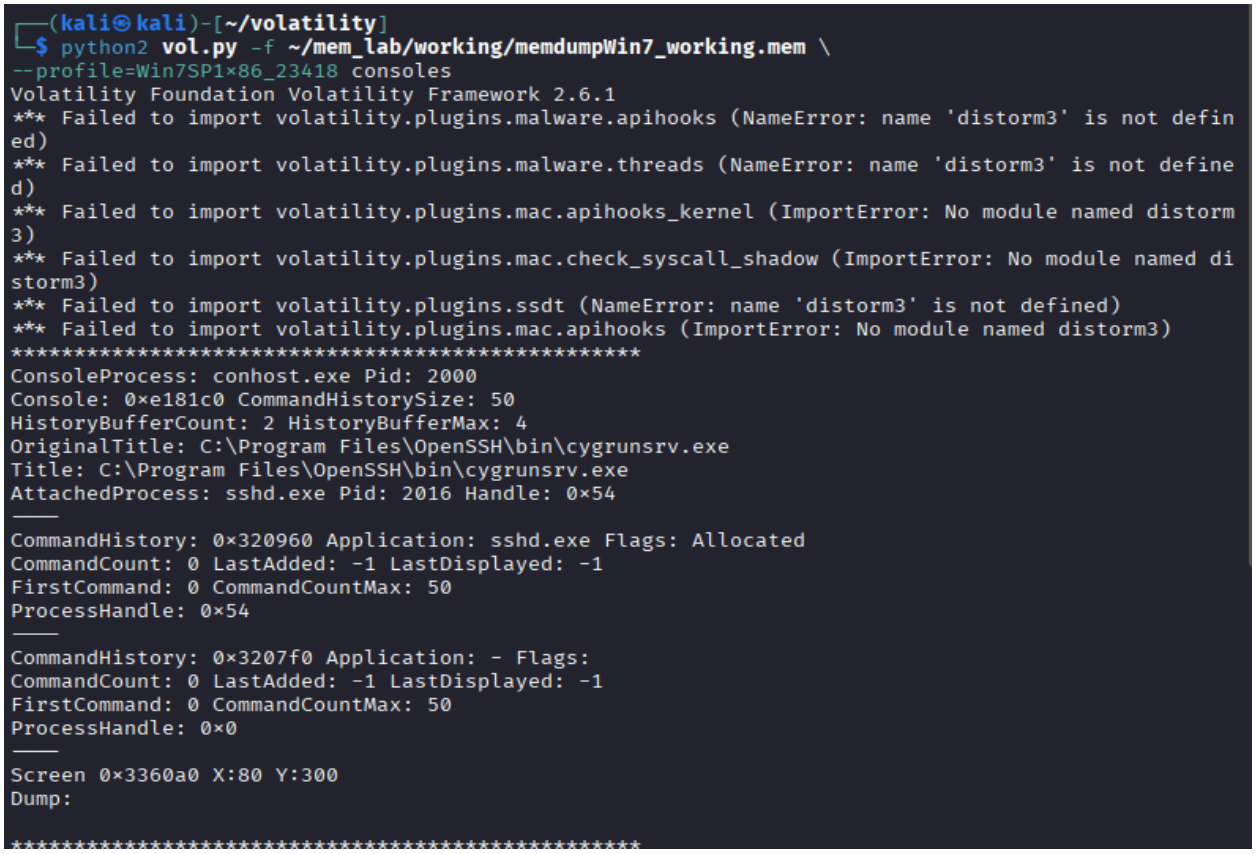
Interpretation

The combined memory artifacts support the conclusion that a command was entered to copy secret_file.docx from the Downloads directory to drive F:, and that the Windows command interpreter displayed a message indicating that one file was copied. Registry evidence associates F: with a General

UDisk USB mass-storage device, while Shellbags provide additional context about the Downloads folder and F: volume. The findings are consistent with a file-copy operation to the USB-associated drive.

Limit

The memory evidence does not independently verify the destination file's contents, integrity, or continued existence. It does not establish the identity of the person operating the system, whether the transfer was unauthorized or unlawful, or the operator's intent. The exact execution time of the copy command has not been established by the recovered console history. Matching image hashes demonstrate that the original and working copies were identical when checked, but do not by themselves establish the image's original authenticity or complete chain of custody.



```
┌──(kali㉿kali)-[~/volatility]
└─$ python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem \
--profile=Win7SP1x86_23418 cmdscan
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defin
ed)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not define
d)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm
3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named di
storm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
**************************************************
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: 3?4?2????
Cmd #37 @ 0x3200c4: 3?4?2????
**************************************************
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????
Cmd #37 @ 0x10cff0: ?????
```

```
┌──(kali㉿kali)-[~/volatility]
└─$ python2 vol.py \
-f ~/mem_lab/working/memdumpWin7_working.mem \
--profile=Win7SP1x86_23418 \
printkey -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2' \
2>&1 | tee ~/mem_lab/outputs/V15_MountPoints2.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defin
ed)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not define
d)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm
3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named di
storm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: MountPoints2 (S)
Last updated: 2019-01-06 15:03:07 UTC+0000

Subkeys:
  (S) CPC
  (S) F
  (S) {421e7e52-11dd-11e9-b3cf-08002709e15d}
  (S) {762f4ebc-60ea-11e5-83af-806e6f6e6963}
  (S) {8358fe6d-60fa-11e5-bb4a-806e6f6e6963}
  (S) {a5b8a980-608c-11e5-a266-806e6f6e6963}
  (S) {a5b8a983-608c-11e5-a266-806e6f6e6963}

Values:
```

```
-f ~/mem_lab/working/memdumpWin7_working.mem \
--profile=Win7SP1x86_23418 \
printkey -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2\F' \
2>&1 | tee ~/mem_lab/outputs/V16_MountPoints2_F.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defin
ed)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not define
d)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm
3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named di
storm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Legend: (S) = Stable   (V) = Volatile

----------------------------------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: F (S)
Last updated: 2019-01-06 15:03:06 UTC+0000

Subkeys:

Values:

┌──(kali㉿kali)-[~/volatility]
└─$ cat ~/mem_lab/outputs/V16_MountPoints2_F.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defin
ed)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not define
d)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm
3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named di
storm3)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Legend: (S) = Stable   (V) = Volatile

----------------------------------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: F (S)
Last updated: 2019-01-06 15:03:06 UTC+0000

Subkeys:
```

6. Timeline Summary

The memory image was identified as a Windows 7 Service Pack 1 system, with the captured system time reported as 2019-01-06 15:09:06 UTC by Volatility's imageinfo output (Appendix B).

The recovered console history contains the commands ipconfig, cd Downloads, and copy secret_file.docx F:. The console screen buffer also contains the message 1 file(s) copied. (Appendix B, cmdscan and consoles evidence). Observed artefact: The command and copy-success message were recovered from memory. Interpretation: This supports that the command interpreter reported a copy operation from the Downloads directory to drive F:. The recovered output does not establish the exact time the command was executed.

The USB storage registry key and the MountedDevices registry output were both last updated at 2019-01-06 15:02:54 UTC. The MountedDevices data associates F: with a USBSTOR device path identifying a General UDisk device (Appendix B). Observed artefact: The registry contains a USB mass-storage entry and a matching device path for F:. Interpretation: This supports the association between the destination drive and a USB storage device; the registry timestamps are not proof of the precise connection or file-copy time.

The MountPoints2 key was last updated at 15:03:07 UTC, while its F subkey was last updated at 15:03:06 UTC. Shellbags records include C:\Users\IEUser\Downloads and F:\; the relevant volume-entry key was last updated at 15:07:03 UTC (Appendix B, V15_MountPoints2.txt, V16_MountPoints2_F.txt, and V17_shellbags.txt). Observed artefact: These registry records contain the F: mount-point and folder/volume information. Interpretation: They provide additional context consistent with Windows recording the USB-associated volume and the Downloads folder. These timestamps describe registry or folder metadata, not a verified file-transfer timeline.

Overall, the recovered console output, USB registry records, mounted-device mapping, and Shellbags are consistent with a command being used to copy secret_file.docx to a USB-associated drive. The available evidence does not establish the exact time of the copy, the identity or intent of the operator, or whether the destination file's contents remained intact. The supporting artefacts and their limitations are documented in Appendix B.

Item	**Assessment**
Confidence level and basis	Moderate confidence that the recovered evidence supports a reported file-copy operation from C:\Users\IEUser\Downloads to F: and that Windows associated F: with a General UDisk USB storage device. This assessment is based on the recovered command history, console screen buffer showing 1 file(s) copied., USBSTOR and MountedDevices registry entries, and Shellbags records. Confidence is lower regarding the identity of the operator, the exact execution time, and whether the transfer was unauthorized.
Limitation 1 (capture timing)	The memory image reports a system time of 2019-01-06 15:09:06 UTC. This is the time reported by the image's metadata, not necessarily the exact time of acquisition or the time of the file-copy operation. The recovered command history does not provide a verified execution timestamp.
Limitation 2 (profile vs person)	The recovered artifacts identify the Windows profile as IEUser. A profile name identifies an account context, not a specific human being. The evidence does not establish who was physically operating the computer when the command was entered
Limitation 3 (command success)	The console screen buffer contains 1 file(s) copied., which supports that the command interpreter reported a successful copy operation. However, the memory evidence reviewed does not independently verify the destination file's contents, integrity, or continued existence.
Limitation 4 (shared access)	The system may have been accessible to more than one person, including anyone with access to the logged-in session. No evidence reviewed conclusively establishes exclusive access to the computer or account at the relevant time.
Limitation 5 (incomplete / overwritten evidence)	Memory captures are snapshots of volatile system state. Relevant command history, processes, file data, or other artifacts may have been absent, overwritten, or not recovered by the available Volatility plugins. The recurring distorm3 import warnings also

	indicate that some plugins were unavailable, so the analysis was not exhaustive.
Alternative explanation(s)	The copy operation may have been legitimate, authorized file transfer or routine data handling rather than unauthorized exfiltration. Another person may have operated the logged-in session. The command and success message could reflect an attempted transfer whose destination contents have not been independently verified. The evidence does not establish the purpose of the transfer or whether the file contained confidential information.

Appendix A. Labelled Screenshots and Volatility Output

All memory-analysis commands used Volatility Framework 2.6.1 with the profile Win7SP1x86_23418 and the working memory image ~/mem_lab/working/memdumpWin7_working.mem, unless otherwise stated. The figure descriptions below identify the evidence each screenshot should show.

Figure	Plugin	Command and profile	Description	Supports (evidence IDs)
A1	imageinfo	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem imageinfo	Shows the memory-image metadata, Windows 7 SP1 identification, image time and suggested profiles.	V1_imageinfo.txt
A2	cmdscan	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 cmdscan	Shows recovered command history, including ipconfig, cd Downloads, and copy secret_file.docx F:.	V5_cmdscan.txt, if saved; otherwise the actual command-history output
A3	consoles	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 consoles	Shows the console history and screen buffer, including 1 file(s) copied.	V6_consoles.txt
A4	printkey	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 printkey -K 'ControlSet001\Enum\USBSTOR'	Shows the General UDisk USB mass-storage registry entry and its last-updated time.	V12_USBSTOR.txt
A5	printkey	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 printkey -K 'MountedDevices'	Shows the F: mapping and the associated USBSTOR device path.	V14_MountedDevices.txt
A6	printkey	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 printkey -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2'	Shows the MountPoints2 key and its subkeys, including F and a volume GUID.	V15_MountPoints2.txt
A7	printkey	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 printkey -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2\F'	Shows the F subkey, its last-updated time, and the absence of listed values or subkeys.	V16_MountPoints2_F.txt

Figure	Plugin	Command and profile	Description	Supports (evidence IDs)
A1	imageinfo	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem imageinfo	Shows the memory-image metadata, Windows 7 SP1 identification, image time and suggested profiles.	V1_imageinfo.txt
A2	cmdscan	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 cmdscan	Shows recovered command history, including ipconfig, cd Downloads, and copy secret_file.docx F:.	V5_cmdscan.txt, if saved; otherwise the actual command-history output
A3	consoles	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 consoles	Shows the console history and screen buffer, including 1 file(s) copied.	V6_consoles.txt
A4	printkey	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 printkey -K 'ControlSet001\Enum\USBSTOR'	Shows the General UDisk USB mass-storage registry entry and its last-updated time.	V12_USBSTOR.txt
A5	printkey	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 printkey -K 'MountedDevices'	Shows the F: mapping and the associated USBSTOR device path.	V14_MountedDevices.txt
A8	shellbags	python2 vol.py -f ~/mem_lab/working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 shellbags	Shows Shellbags records for C:\Users\IEUser\Downloads and F:\, with the relevant recorded metadata.	V17_shellbags.txt